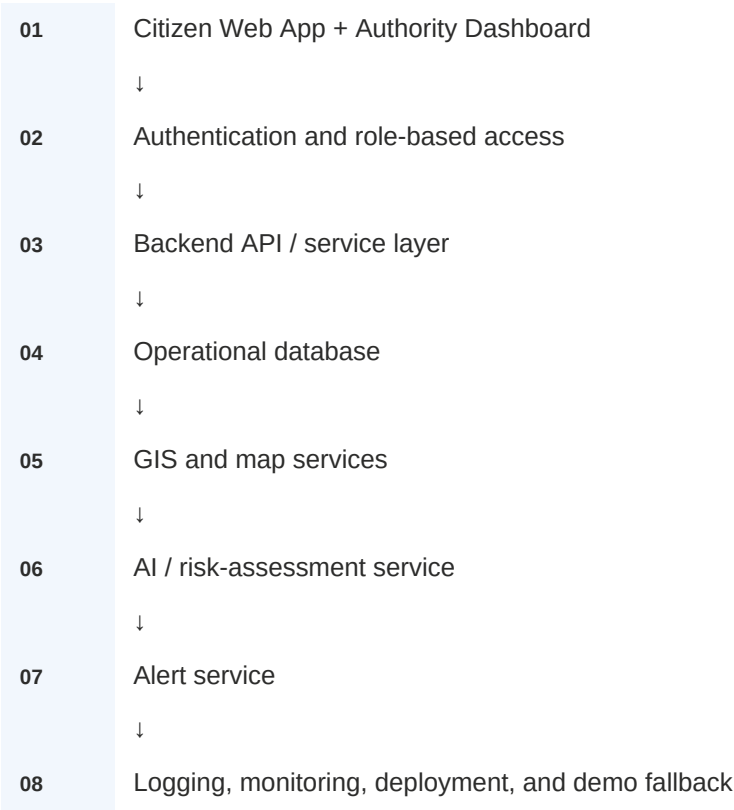


09 SYSTEM ARCHITECTURE

System Architecture

A technology-neutral view of TRINETRA’s components, data domains, boundaries, and ownership.

Logical system architecture



Component responsibilities

Component	Primary responsibility	Owner
Citizen application	Dashboard, alerts, map, safe zones, resources, reporting	Abhinav
Authority application	Incident/report monitoring, alert publishing, resource management	Abhinav
Backend API	Business rules, validation, RBAC, integration contracts	Abhinav
Database	Users, incidents, reports, alerts, areas, resources, risk results	Abhinav
GIS layer	Spatial formats, map layers, bounds queries, freshness	Rahul
Risk service	Bounded advisory assessment, version, explanation, fallback	Prinshu
Operations	Testing, security, deployment, monitoring, recovery	Prinshu

Core data entities

- User and role
- Disaster / incident
- Citizen report
- Affected area
- Alert
- Safe zone
- Emergency resource
- Risk assessment result

Architecture decisions to freeze

- Technology stack and environment
- Authentication/session model
- API schema and error standard
- Geospatial format and coordinate system
- Data provenance and freshness model
- Deployment, observability, security, backup, and fallback